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Time-averaged field at the equator: Results from Ecuador

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[1] Seventy sites were collected from Pliocene to recent lavas in Ecuador between 0.19° and 1.1°S latitude. Lightning affected many of the sites sampled, and the final data set consisted of 51 sites with α_{95} of 10° or less; 21 flows yielded normal directions of magnetization (declination = 354.6°, inclination = −5°, α_{95} = 7.2°) and 30 yielded reverse (declination = 183.5°, inclination = 5.6°, α_{95} = 5°). These sites pass a reversal test at a high level. The combined data give a direction of declination = 359.9°, inclination = −5.4°, and α_{95} = 4.2°, which is just significantly different from the GAD field but is in agreement with a GAD field plus 5% quadrupole field. Mean VGPs are not significantly different from the geographic poles. The ASD for the combined sites is 13.3°, which is in agreement with model "G." Paleointensity measurements yield values of VADM, which range from 1.39×10^{22} to 22.6×10^{22} Am^2. The mean value for the reverse field is 3.2×10^{22} Am^2, while for the normal field it is 10.2×10^{22} Am^2. The ^{40}Ar/^{39}Ar dates on selected sites show that they are 2.6 Ma and younger. The age dates and magnetic polarity indicate that the Chacana structure is Brunhes/Matuyama in age.

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1. Introduction

[2] Earth's magnetic field originates in its core and is observed at the surface where it may be studied at magnetic observatories and in recent years by satellite. The longest magnetic observatory records of the magnetic field reach several hundred years into the past. The distribution of the field over the surface of Earth since the 16th century has been mapped from records kept on sailing ships [*Bloxham et al.*, 1989]. The history of the field for times older than the 15th century requires the paleomagnetic study of suitable earth materials. Modeling of Earth's magnetic field has recently had some success [*Glatzmaier and Roberts*, 1995] and it has also been possible to use surface observations to model the field at the core-mantle boundary. These studies have indicated that some features of the magnetic field have persisted for several million years, such as flux lobes at high latitude [*Bloxham and Jackson*, 1992; *Johnson and Constable*, 1995]. The success of these modeling efforts rests on the quality and quantity of the available data.

[3] It became apparent by 1997, that the paleosecular variation in lavas (PSVL) database for the last 5 Ma of Earth's history available for use by the modeling community was inadequate [*McElhinny and McFadden*, 1997]. Data were compiled by several authors, beginning with *McElhinny and Merrill* [1975] and updated by *Lee* [1983], *Quidelleur et al.* [1994], and *Johnson and Constable* [1996]. *McElhinny and McFadden* [1997] argued that many of the older studies should be redone using modern analytical techniques since it was clear that many older analyses involved data which were not adequately cleaned of overprints; this is particularly true for the data recording reverse directions of the magnetic field.

[4] The response to this clear deficiency in the data set by the scientific community has been an attempt to upgrade the data to meet modern standards. This has resulted in several new studies, which have attempted to expand and upgrade the database [*Carlut et al.*, 2000; *Tauxe et al.*, 2000]. This has also brought into being the Time Averaged Field Initiative (TAFI), a consortium of four universities (Scripps Institution of Oceanography, University of Florida, University of Massachusetts, and the University of Alaska) with the intent to expand the existing data set and to redo and upgrade some strategically placed localities to modern standards. This program has been successful and new results have been obtained from Nunivak Island (on a collection originally made by Allen Cox); a restudy of the Aleutian Islands [*Stone and Layer*, 2001]; British Columbia, Canada; the Snake River Plain; the San Francisco Peaks area of Arizona; the Trans-Mexican Volcanic Belt; Costa Rica; Chile; Patagonia; and another restudy of samples collected by Cox [*Mankinen and Cox*, 1988] from Antarctica. These new results provide data from the Arctic to the Antarctic at roughly every 10° of latitude down the backbone of the Americas. Other localities were sited to fill in gaps at high latitudes in order to study the persistence of flux lobes. The study reported here presents new data from the equator, filling a gap in the latitudinal coverage which was omitted in the earlier program. It is important to study the time averaged field at the equator since the quadrupole field is most easily detected at that latitude.

[5] Data from two previous studies have been published from rocks collected from equatorial latitudes both from the Galapagos Islands which lie just south of the equator 1000 miles to the west of Ecuador proper in the eastern Pacific Ocean. *Cox* [1971] published data from San Cristobal Island on 17 flows which gave excellent results even though for many of the sites only NRM data were reported. The α_{95} of these sites were all 5° or less and it is very probable that these results accurately reflect Earth's magnetic field at the time that the lavas cooled. The site mean direction is not significantly different from the geocentric axial dipole. The second study of the Galapagos Islands was reported by *Rochette et al.* [1997] in which only the data concerning the angular standard deviation of the VGP distribution were presented. The samples were subject only to blanket alternating field (AF) demagnetization (the application of a single, presumably optimal demagnetization level, which does not allow one to account for differential demagnetization behavior on a specimen level). Moreover the site data was not presented and no site information is available in the paper.

2. Geology

[6] Ecuador is ideally placed for a study of the paleosecular variation of lava flows. Ecuadorian volcanoes form part of the northern segment of the Andean Volcanic Belt, called the Northern Volcanic Zone (NVZ), which extends from 5°N to 2°S latitude (covering parts of Colombia and Ecuador). From north to south the NVZ begins as a single row of stratovolcanoes that widens into three main

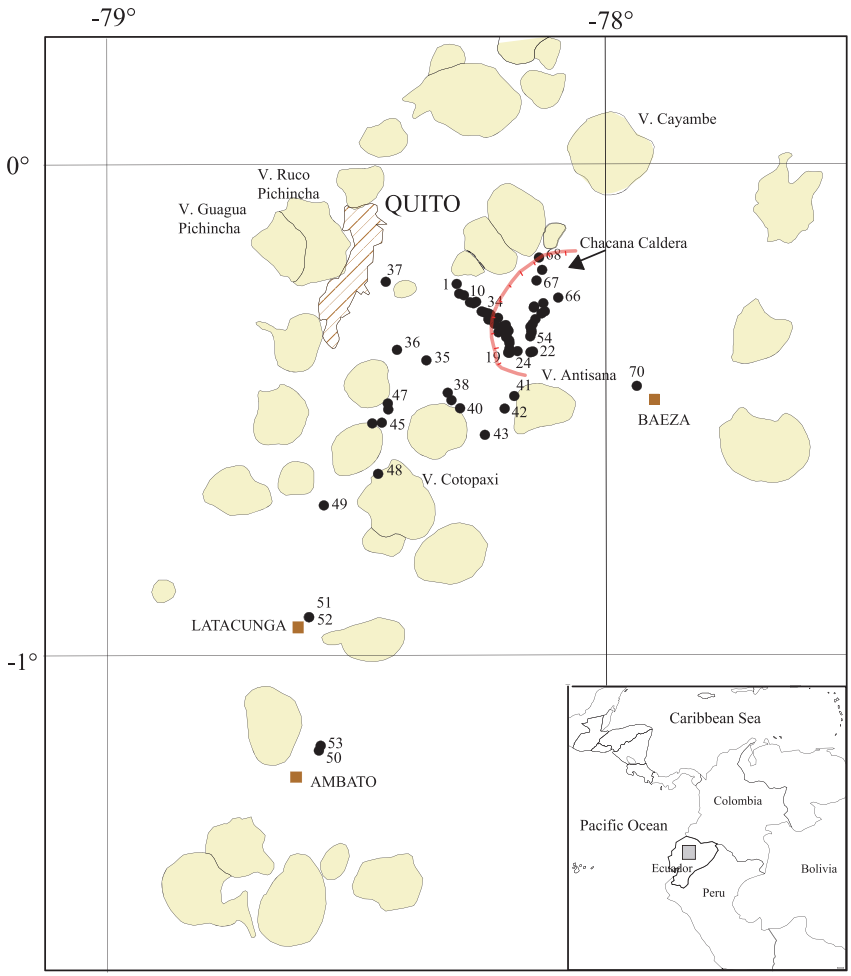


Figure 1. Map of area of study in Ecuador showing site locations (circles). Plio-Quaternary stratovolcanoes (yellow areas) are also shown. Geologic features were obtained from the geologic map of Ecuador by *Litherland and Egüez* [1993].

rows at about 2°N (the political boundary between Colombia and Ecuador). These are the Volcanic Front to the west, the Cordillera Real to the east, and the volcanoes of the Interandean Depression (between the first two). Widening in the NVZ was accompanied by enhanced volcanism that produced more than 54 stratovolcanoes which are attributed to the subduction of the Carnegie Ridge riding upon the Nazca Plate [*Hall and Wood*, 1985; *Barberi et al.*, 1988]. Subduction began at about 5 Ma [*Pennington*, 1981]. The present chain of stratovolcanoes (Figure 1) range in age from 2.5 Ma to the present, and range from calcalkaline basaltic andesites to rhyolites in compositions [*Hall and Calle*, 1982; *Barberi et al.*, 1988]. Eleven volcanoes in Ecuador are still active and three erupted within the last two years. Several lava flows have documented historic eruption dates [*Hall*, 1977]. Figure 1 shows the distribution of volcanoes in northern Ecuador; many have isotopic dates indicating that the lavas are late Pliocene or younger.

3. Sampling

[7] The sampling trip to Ecuador took place during June 2003. Samples were collected using a hand-held diamond drill that produces cores 2.5 cm in diameter and several centimeters in length. Most samples were taken from unaltered massive lavas in well defined flow units. The samples were oriented using a magnetic compass and when possible a sun compass. Cloud cover was frequent during the sampling, so some sites do not have sun bearings. Samples were taken preferentially from road cuts, quarries, or from the base of cliffs in order to avoid the effects of lightening strikes.

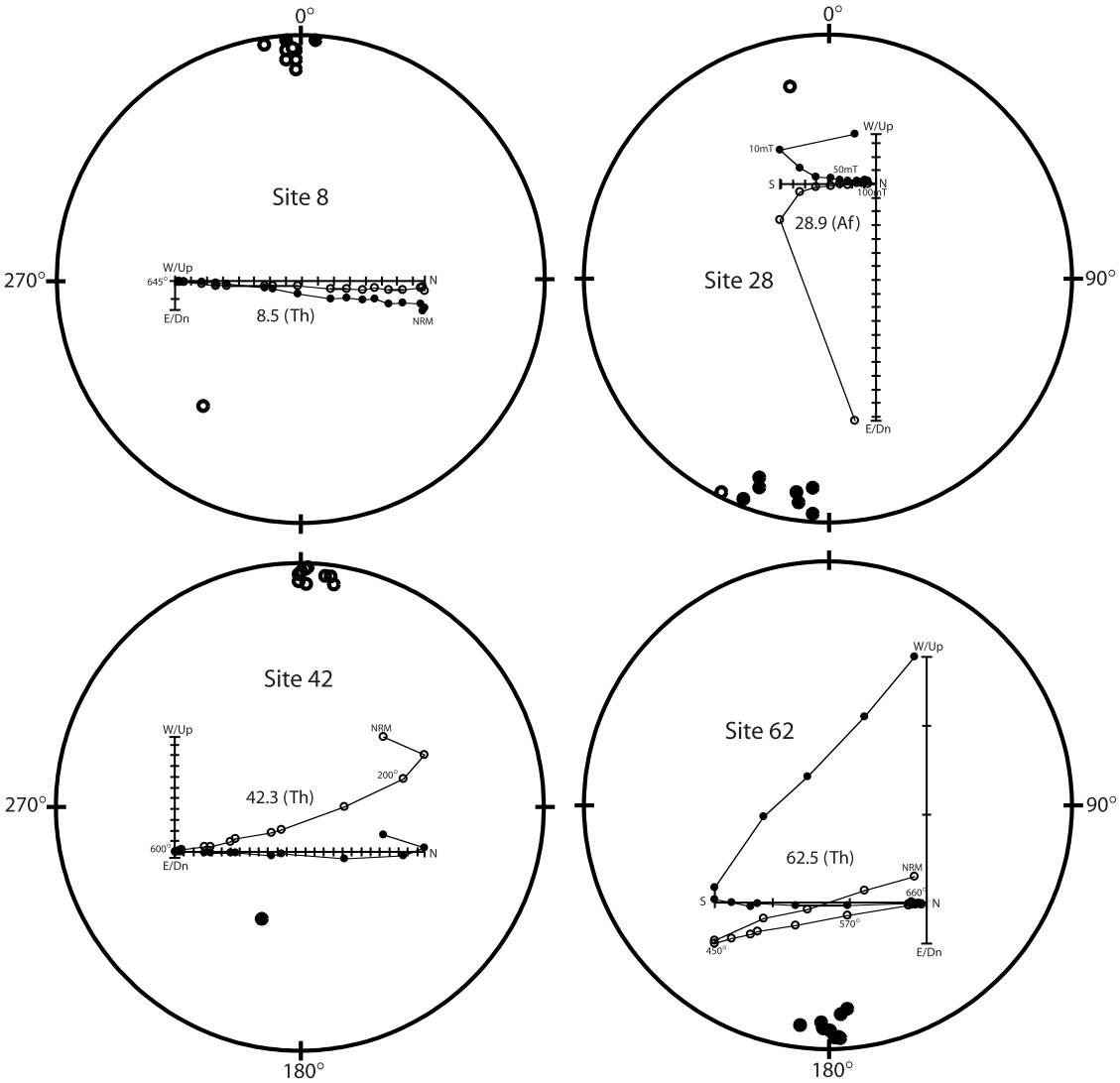


Figure 2. Stereographic plots of sample directions from sites 8, 28, 42, and 62.Three of the sites have single sample outliers which were not included in the site statistics. Demagnetograms are shown of typical samples from each site. The solid circles indicate the vector end point of the declination, and open circles indicate the inclination above or below the horizontal. Intensity is given A/m. Sites 8, 42, and 62 were thermally demagnetized, in some cases at temperatures up to 660°C, while site 28 was demagnetized using alternating fields up to 100 mT.

Sampling was carried out over several meters to lessen the chance that multiple samples could be affected by the same lightning strike and ten samples were taken from every site (lava).

[8] Most of the region that was sampled lies in the Cordillera Real to the east of Quito and the Inter Andean valley. Many sites were located along the road from Pifo to Baeza, which passes through the village of Papallacta. The road bisects a large caldera that has been termed the Chacana complex by *Hall and Mothes* [1997]. The oil pipeline from the Amazon basin follows this route and the region is the source of much of the drinking water for Quito. Several younger volcanoes are associated with this structure such as the Antisana volcano which we sampled. Sampling was also carried out at the active volcano Cotopaxi and in the Pillaro area. Figure 1 shows the distribution of the seventy sampled sites.

4. Laboratory Studies

[9] Samples were returned to Gainesville where they were sliced into specimens 1.25 cm in length. Different length to diameter ratios had no discern-

Table 1. Paleomagnetic Statistics[a]

Site					Mean Direction				VGP		
ID	Longitude, °W	Latitude, °S	n/N	P	Dec	Inc	α_{95}	κ	Latitude, °N	Longitude, °E	Age ± 2σ, Ma
1	78.30	0.24	10/10	N	336.3	0.3	4.5	118	66.3	192.6	
2	78.24	0.30	13/19	N	330.7	10.4	6.3	44	60.2	202.9	
3	78.24	0.30	4/10	I	327.5	11.2	12.3	57	57.0	202.7	
5	78.22	0.31	8/10	N	345.1	6.5	4.6	147	74.7	205.4	
6	78.21	0.33	11/11	R	176.8	12.6	4.8	93	−83.2	309.6	1.89 ± 0.10
7	78.20	0.33	8/10	R	211.0	1.1	9.1	38	−59.0	192.3	
8	78.20	0.33	9/10	N	357.9	−5.6	3.9	176	86.7	142.0	0.45 ± 0.08
9	78.29	0.36	7/10	R	171.2	16.6	4.7	168	−78.0	328.3	
10	78.28	0.26	8/10	I	259.9	−17.8	10.7	28	−9.9	182.4	
11	78.22	0.32	11/12	N	10.3	−8.5	1.7	733	79.0	32.9	
12	78.21	0.34	10/10	I	233.2	−10.2	6.4	58	−36.6	185.1	
13	78.21	0.33	9/9	I	220.4	−14.3	3.5	213	−49.0	180.3	1.51 ± 0.24
14	78.26	0.28	10/10	R	164.8	10.9	2.6	360	−74.0	352.5	1.46 ± 0.05
15	78.26	0.28	11/11	R	159.0	8.5	6.1	57	−68.6	0.7	
16	78.20	0.33	11/11	R	198.6	21.0	4.8	90	−68.7	222.1	2.59 ± 0.47
17	78.20	0.35	10/10	R	176.1	2.8	5.7	72	−86.0	356.7	2.15 ± 0.19
18	78.19	0.36	10/10	N	347.7	−29.2	3.6	181	70.5	139.7	
19	78.20	0.38	10/10	R	198.3	17.5	3.7	167	−69.9	217.5	
20[b]	78.15	0.38	9/10	N	348.0	−6.5	11.6	21	77.7	178.2	
21	78.15	0.38	6/9	R	181.3	23.0	3.3	405	−78.3	275.6	
22	78.15	0.38	7/10	R	169.7	−5.2	7.6	64	−79.3	28.1	
23	78.18	0.38	10/11	R	177.6	−0.9	5.3	84	−87.5	30.9	
24[b]	78.19	0.38	6/10	R	191.6	3.6	11.7	34	−78.3	198.9	
25	78.19	0.34	10/10	R	185.7	11.7	1.9	647	−82.0	236.3	
26	78.20	0.35	11/12	R	201.6	2.3	6.4	52	−68.4	194.0	
27	78.19	0.36	9/10	N	349.4	−27.7	2.7	368	72.2	137.5	0.44 ± 0.15
28	78.19	0.37	6/10	R	190.4	8.8	7.1	91	−78.8	213.3	
29	78.23	0.31	9/10	R	197.9	−6.3	3.4	230	−71.8	180.7	2.71 ± 0.19
30	78.23	0.31	11/11	R	195.5	−3.2	2.0	540	−74.4	184.7	
31	78.23	0.31	9/10	R	195.8	−1.8	2.5	414	−74.2	187.4	
32	78.27	0.28	6/10	R	172.8	−0.8	4.3	239	−82.8	17.1	
33[b]	78.26	0.28	6/11	R	178.7	−4.1	12.1	32	−87.3	72.6	
34	78.25	0.30	10/11	R	198.8	6.8	4.4	122	−71.0	201.4	
35	78.36	0.40	10/10	N	8.0	9.9	2.4	419	80.4	337.6	
36	78.42	0.38	9/10	N	351.7	−5.5	5.3	94	81.4	175.5	
37[b]	78.44	0.24	7/10	R	168.4	−18.7	22.6	8	−74.8	52.3	
38	78.32	0.46	9/10	N	14.5	−24.8	2.9	308	70.9	53.4	
39	78.31	0.48	8/10	N	357.3	−1.5	8.4	45	87.3	185.9	
40	78.29	0.49	10/10	R	189.9	11.3	3.6	180	−78.8	219.7	
41	78.18	0.47	10/10	R	173.6	6.7	5.8	71	−83.0	347.4	
42	78.20	0.50	9/10	N	3.2	−6.8	2.9	318	85.7	54.2	
43	78.24	0.55	8/8	N	13.4	−12.8	3.0	352	75.4	36.0	
44	78.47	0.53	4/10	N	339.3	−8.2	9.4	96	69.0	181.4	
45	78.45	0.52	6/10	N	352.3	−9.1	8.6	62	81.3	163.6	
47	78.44	0.48	10/10	N	6.6	2.3	4.0	144	83.2	357.6	1.33 ± 0.30
48	78.46	0.63	10/10	N	351.5	−1.8	6.0	66	81.5	189.7	
49	78.56	0.69	10/10	N	357.4	−4.6	6.0	67	86.9	159.6	
50	78.57	1.19	8/10	R	188.2	8.2	2.9	360	−81.3	211.3	
51	78.59	0.92	10/10	R	181.9	−2.2	4.0	147	−87.2	144.7	1.67 ± 0.19
52	78.59	0.92	9/10	R	173.3	−13.3	1.7	907	−79.8	60.4	
53	78.57	1.18	10/10	R	173.3	−2.3	3.0	255	−82.9	30.6	
56[b]	78.15	0.34	6/10	R	145.7	85.6	38.3	4	−7.6	286.8	
57	78.15	0.33	8/10	R	176.4	11.4	6.3	79	−83.5	315.3	
59[b]	78.15	0.32	7/10	R	167.0	−16.2	29.4	5	−74.5	45.7	
60	78.14	0.31	9/10	I	152.9	62.6	4.6	129	−40.1	307.3	
61	78.14	0.29	8/10	R	178.7	10.4	8.5	44	−84.9	296.5	
62	78.14	0.29	9/9	R	179.2	9.7	3.5	217	−85.3	291.7	2.44 ± 0.20
63	78.13	0.30	10/10	R	190.2	12.5	4.2	134	−78.2	222.7	
66	78.10	0.27	9/10	N	355.2	−6.5	3.8	182	84.3	159.9	
67	78.14	0.23	10/10	N	352.8	−12.6	4.2	136	80.5	151.2	
68	78.13	0.19	9/10	N	345.4	33.5	9.8	28	66.6	244.8	

Table 1. (continued)

Site					Mean Direction				VGP		
ID	Longitude, °W	Latitude, °S	n/N	P	Dec	Inc	α_{95}	κ	Latitude, °N	Longitude, °E	Age ± 2σ, Ma
69	78.13	0.21	10/10	R	177.7	−14.3	2.6	346	−82.2	84.9	1.78 ± 0.20
70[b]	77.94	0.45	5/10	N	23.8	−27.2	21.0	14	62.6	43.8	0.21 ± 0.03
Normal sites			21	N	354.6	−5.0	7.2	20	84.2	170.0	
Reverse sites			30	R	183.5	5.6	5.0	28	−85.8	226.5	
All sites			51		359.9	−5.4	4.2	23	87.7	106.0	

[a] Columns indicate, from left to right, site identification (ID); site coordinates in longitude west and latitude south; the number of samples included in the site statistics (n) and the number of samples collected (N); the polarity of the site (P) indicated by N (normal), R (reverse) and I (transitional); declination (Dec), and inclination (Inc) derived from Fisher analysis of the site directions where α_{95} is the radius of the 95% confidence interval and κ is the concentration parameter; VGP coordinates given in latitude north and longitude east; ages of dated sites are given along with 2σ error. Sites 46, 54, 55, 58, 64, and 65 did not yield results because principal components could not be satisfactorily resolved and are omitted from the table. Sites 2 and 4 sampled the same flow and are combined as site 2.
[b] Sites with α_{95} greater than 10°.

ible effect on the results. The NRM of the specimens was measured and AF demagnetization was carried out for those sites which showed evidence (i.e., abnormally high NRM intensity and scattered directions) of having been affected by lightning. Sites in which the NRM was well grouped were demagnetized using thermal demagnetization. Line fitting techniques [*Kirschvink*, 1980] were used to determine the direction of magnetization; usually three or more points were used to determine the line (Figure 2). *Fisher* [1953] statistics were performed on sites that had 4 or more successfully demagnetized samples. Sites occasionally showed well grouped directions with a specimen direction wildly aberrant and such specimens were not included in the analysis (Figure 2). The reason for this aberrant behavior is in some cases lightning or that a sample may have been taken by mistake from a rolled boulder or misoriented. Sites 46, 54, 55, 58, 64 and 65 did not yield results because principal components which deliniates the direction of magnetization could not be determined. The results of this analysis are presented in Table 1. Sites with α_{95} greater than 10° (7) were not

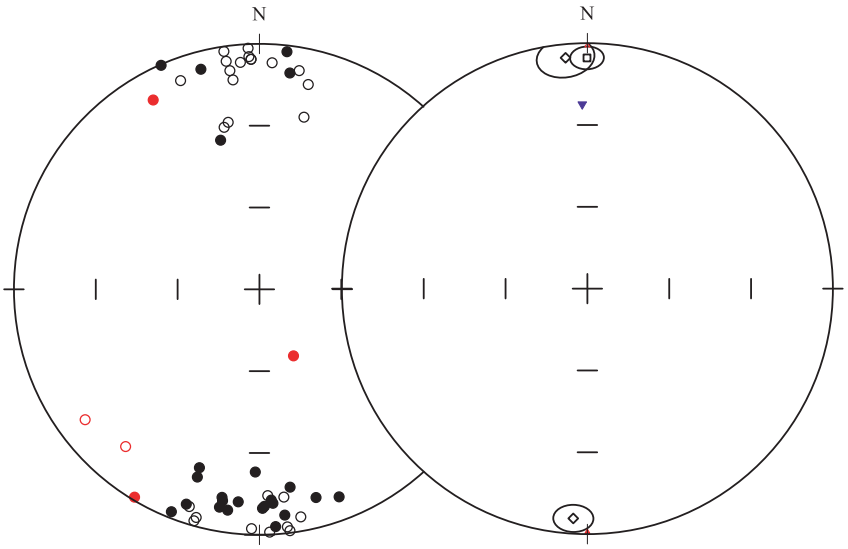


Figure 3. Paleomagnetic directional results from Ecuador. Solid (open) symbols represent directions above (below) the horizontal. (left) Site mean directions of normal, reverse (black circles), and intermediate sites (red circles) with $\alpha_{95} \leq 10°$. (right) Mean directions among the normal and reverse sites (diamonds) and among the overall sites (square) surrounded by the 95% confidence angle (α_{95}). Red triangles pointing up are the expected GAD in the area for normal and reverse fields; a blue triangle pointing down is the 2003 IGRF. Sites were interpreted to have intermediate directions when VGP latitude was less than 62° as calculated from *Vandamme* [1994] for the latitude of study, but sites 2 and 7 were included for calculations of mean directions among sites (see text).

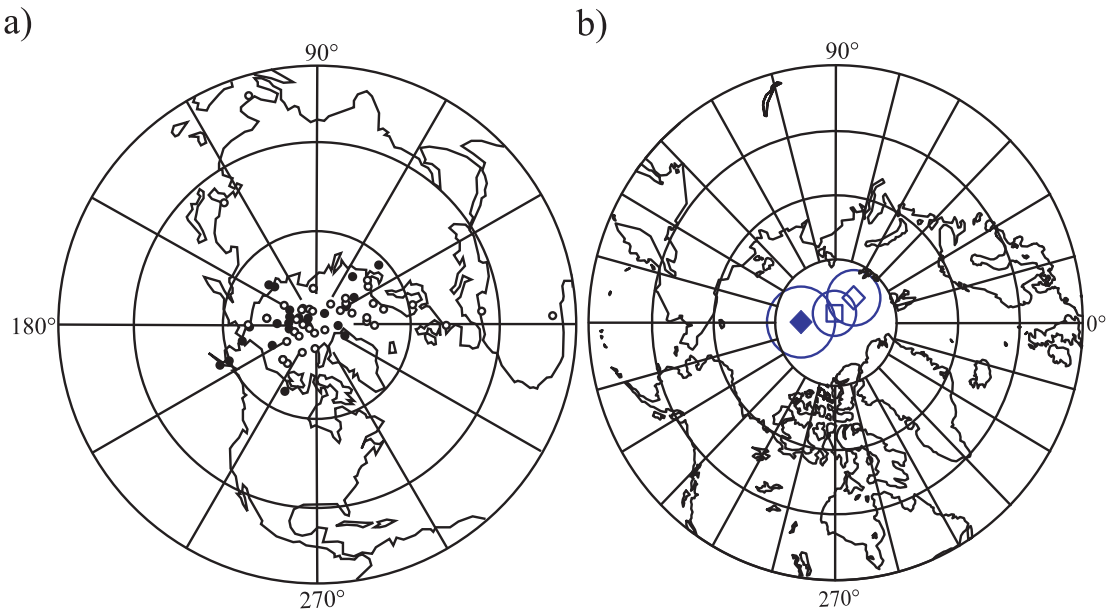


Figure 4. VGPs of data from Ecuador. (a) Site VGPs. Solid/open circles are poles that fall in the Northern/Southern Hemisphere. (b) Mean VGP for global data (open square), the normal data (solid diamond), and reverse data (open diamond) that were calculated excluding sites with $\alpha_{95} > 10°$ and intermediate directions.

considered further and transitional sites (5) were identified using the criterion developed by *Vandamme* [1994]. However, this criterion is very restrictive and would eliminate sites with VGPs below 61° of latitude. Sites 2 and 7 have VGP latitudes of 59° and 60°; we chose to include these sites in the selected data set since they were so close to the cut off latitude. Five sites are regarded as transitional (3, 10, 12, 13, and 60) and eliminated from further study. This left a total of 51 sites to be used in further analysis. Twenty one had normal magnetization and thirty reverse (Figure 3).The normal directions yielded a mean direction D = 354.6°, I = −5.0° with an α_{95} of 7.2°, which is not significantly different from the geocentric axial dipole (D = 360°, I = −0.8°). The mean reverse direction was D = 183.5°, I = 5.6°, α_{95} = 5.0°. This direction is just significantly different from the GAD field. The reverse and normal data have overlapping circles of confidence and pass a reversal test at the "B" level of *McFadden and McElhinny* [1990]. This allows the data to be combined and yields an overall mean direction of D = 359.9°, I = −5.4° with an α_{95} = 4.2° which is significantly different from the geocentric axial dipole.

[10] VGPs were calculated for each site and the distribution shown in Figure 4 with the reverse sites inverted. There is no significant difference between the mean directions among normal and reverse sites and there is no significant difference between the VGP means and the geographic poles. In addition, there is a slight far sided effect, but no right handed bias observed. The VGP scatter is 13.3° with error limits of 11.6° and 15.4° (without considering sites 2 and 7) which is in excellent agreement with model "G" of *McFadden et al.* [1988]. It should be noted that the directions have a Fisherian distribution whereas the VGPs have an elongated distribution normal to the site longitude. This is not what is expected from a magnetic field model recently proposed by *Tauxe and Kent* [2004]. Larger data sets may be needed to adequately test this model.

5. Paleointensity

[11] Paleointensity experiments using the *Thellier and Thellier* [1959] method as modified by *Coe* [1967] were made on 33 replicates of the samples run for directional studies. The samples were selected based on their stability of remanence, in an attempt to improve the relatively low success rate observed in many studies of paleointensity in lavas.

[12] The experiments were run in the paleomagnetic laboratory at the University of Florida, using a MMTD80 thermal demagnetizer specially designed for paleointensity experiments. The ex-

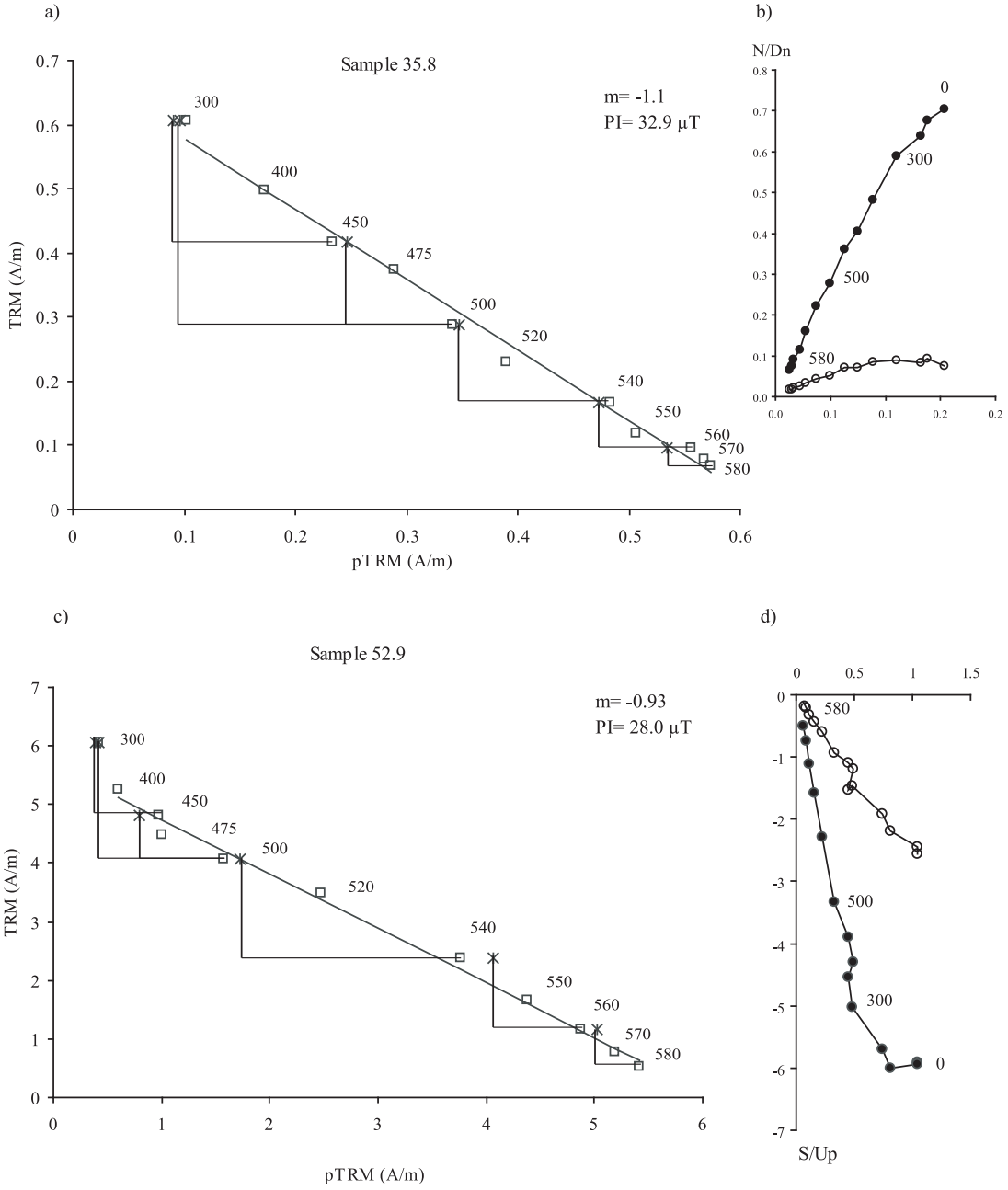


Figure 5. Successful paleointensity results of two samples from sites 35 and 52. (a and c) Arai plots with (b and d) the corresponding Zijderveld plots of the field-off steps to the right. Temperature steps are in °C.

periment consisted of a series of 11 stepwise double heating and cooling steps in a 0 field and a known field of 30 μT. Likewise 5 pTRM checks were performed (as shown in Figure 5). Despite the apparent simplicity of the Thellier method, or any other method to obtain "absolute" paleointensity values, achieving good results has proven to be very difficult and is a matter of controversy. In principle this experiment is only applicable in the case of rocks with single domain grains [e.g., *Dunlop et al.*, 2005] and the extent of the errors incurred upon applying the method in rocks with PSD has not been determined yet.

[13] Paleointensity data were interpreted by analyzing both Arai plots and Zijderveld plots from the off-field steps of the Thellier experiment. The observation of Zijderveld plots is important to

Table 2. Paleointensity Results from Ecuador[a]

Sample	Latitude	Longitude	N	T1, °C	T2, °C	σ_b	σ_b/B	f	g	q	DRATS, %	F, μT	VADM 10^{22} Am^2	P
11.8	−0.32	−78.22	11	300	580	0.10	0.11	0.59	0.88	4.68	14.45	27.65	7.15	N
19.10	−0.38	−78.20	11	300	580	0.03	0.08	1.00	0.84	10.50	17.46	11.33	2.93	R
25.2	−0.34	−78.19	10	400	580	0.02	0.08	0.85	0.84	8.44	8.45	6.01	1.55	R
29.9	−0.31	−78.23	11	300	580	0.01	0.06	0.78	0.89	12.31	9.25	6.46	1.67	R
30.7	−0.31	−78.23	11	300	580	0.02	0.04	0.73	0.87	14.13	9.92	14.41	3.73	R
31.6	−0.31	−78.23	11	300	580	0.02	0.10	0.86	0.88	7.31	25.19	5.38	1.39	R
32.5	−0.28	−78.27	11	300	580	0.03	0.04	0.89	0.81	16.67	1.73	18.06	4.67	R
35.5	−0.40	−78.36	11	300	580	0.04	0.03	0.77	0.87	19.69	4.54	32.94	8.52	N
38.7	−0.46	−78.32	11	300	580	0.10	0.08	0.87	0.88	9.97	12.64	38.61	9.98	N
43.6	−0.55	−78.24	8	300	580	0.02	0.06	0.82	0.81	11.34	7.39	10.17	2.63	N
52.9	−0.92	−78.59	10	400	580	0.03	0.03	0.81	0.86	25.05	8.19	27.95	7.22	R
60.5	−0.31	−78.14	10	400	580	0.06	0.07	0.35	0.86	4.44	7.77	27.31	7.06	I
62.7	−0.29	−78.14	10	400	580	0.03	0.08	0.88	0.85	9.05	0.74	11.06	2.86	R
66.1	−0.27	−78.10	7	500	580	0.11	0.04	0.51	0.81	10.52	−25.98	87.48	22.62	N

[a] N is the number of temperature steps used to calculate the value of paleointensity in the interval of temperatures T1 and T2 (°C). Quality parameters defined by *Coe et al.* [1978] are the standard error of the slope (σ_b), the standard error of the slope divided by the slope (σ_b/B), the fraction of the NRM used for calculation of paleointensity (f), the gap factor (g), and the quality factor (q). DRATS is the pTRM difference ratio sum, F is the value of paleointensity, VADM is the virtual axial dipole moment, and P is the magnetic polarity of the site.

ensure that the segment of unblocking temperatures used to calculate a paleointensity value trends to the origin and to observe any changes in the direction caused by magnetomineralogic changes during heating.

[14] Paleointensity data was interpreted from results that displayed linearity of the Arai plot and had good pTRM checks. On the basis of the finding by *Dunlop et al.* [2005] that pTRM checks can be unsuccessful when using multidomain (MD) materials, even if no chemical alteration has taken place, we considered it unsafe to make paleointensity interpretations from two-slope Arai plots that apparently had an interpretable segment with successful pTRM checks. Curved Arai plots indicating MD behavior were also excluded from analysis.

[15] There is a lack of consensus about the quality criteria and cutoff values that should be used for data selection. *Coe et al.* [1978] established several quality factors: the standard error of the slope in the Arai plot (σ_b), the fraction (f) of the total (extrapolated) NRM that is used for paleointensity calculations (ΔY/a), the gap factor (g) that quantifies how evenly distributed are the NRM values in the slope, and a quality factor (q) that combines the previous three factors (q = b * f g/σ_b). *Coe et al.* [1978] did not set up a cutoff value for any of these quality factors and we adopted the value of q > 1 that was used by *Selkin and Tauxe* [2000].

[16] One of the main quality criteria is the coincidence of pTRM checks [*Coe*, 1967] with the initial measure of pTRM. *Coe* [1967] did not set up any quantitative control to consider the success of a pTRM check. *Pick and Tauxe* [1993] using submarine basaltic glass (SBG) established that pTRM checks should coincide within 5% of the initially measured pTRM. This quality standard is rarely met when using materials such as lava flows. A variety of alternatives have been implemented to analyze pTRM checks [e.g., *Juarez and Tauxe*, 2000; *Selkin and Tauxe*, 2000; *Tauxe and Staudigel*, 2004]. In this study the approach by *Tauxe and Staudigel* [2004] was used, in part because it appears to be more practical. These authors defined a parameter called "DRATS" (pTRM difference ratio sum), which is the sum of the differences between the initially measured pTRM and the pTRM checks, normalized by the maximum pTRM used for paleointensity calculations. By doing this sum, the negative and positive differences between initial pTRM and pTRM checks tend to cancel out, which according to *Tauxe et al.* [2004] is advantageous because it gives relevance to trends indicating mineralogic changes rather than to scatter brought about from experimental conditions. The maximum value of "DRATS" considered as a successful result was 26%. A value of 25% was used by *Tauxe and Staudigel* [2004] and even a higher value of 30% was used by *Tauxe et al.* [2004] when analyzing paleointensity data on lava flows. For the calculation of DRATS the second pTRM check at 300° was not taken into consideration because it covers unblocking temperature ranges accounted for in other pTRM checks. The results of this analysis are given in Table 2.

[17] The success rate in this study is about 45%, which is much better than we have obtained in other studies on lavas. We obtained acceptable results from fourteen sites, 5 normal, 8 reverse and one

Table 3. (Representative Sample). $^{40}Ar/^{39}Ar$ Analytical Data[a] [The full Table 3 is available in the HTML version of this article at http://www.g-cubed.org.]

Temperature, °C	40(r)/39(k)	±1 SD for 40(r)/39(k)	40(r + a)	±1 SD for 40(r + a)	$^{40}Ar/^{39}Ar$	±1 SD for $^{40}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	±1 SD for $^{37}Ar/^{39}Ar$	$^{36}Ar/^{39}Ar$	±1 SD for $^{36}Ar/^{39}Ar$	^{40}Ar, mol	^{40}Ar*, %	Cumulative ^{39}Ar, %	Calculated Age, Ma	Error in Age (±2 SD)	K/Ca
EC-6[b]																
550	49.411	4.873	23.851	0.126	176.830	2.060	0.6789	0.0599	0.4314	0.0167	2.385E-16	27.93	6.85	43.16	8.41	0.633
550	67.165	8.131	22.242	0.083	178.611	1.578	0.5923	0.0504	0.3773	0.0275	2.225E-16	37.59	6.33	58.42	13.92	0.726
650	2.526	0.160	0.935	0.018	5.547	0.114	0.8422	0.0305	0.0104	0.0004	9.395E-18	45.51	8.60	2.23	0.28	0.510
750	2.146	0.065	1.599	0.021	3.464	0.057	0.6127	0.0231	0.0045	0.0001	1.611E-17	61.92	23.62	1.90	0.11	0.701
835	2.008	0.092	1.078	0.019	2.662	0.053	0.5726	0.0212	0.0023	0.0003	1.089E-17	75.42	20.78	1.77	0.16	0.751
935	2.229	0.073	0.678	0.017	2.634	0.069	0.6508	0.0252	0.0015	0.0001	6.847E-18	84.58	13.20	1.97	0.13	0.660
1030	3.159	0.207	0.446	0.017	4.065	0.158	0.8234	0.0363	0.0032	0.0005	4.486E-18	77.67	5.60	2.79	0.36	0.522
1160	3.908	0.274	0.482	0.017	6.564	0.237	1.7013	0.0629	0.0094	0.0005	4.842E-18	59.48	3.74	3.45	0.48	0.252
1400	23.421	4.143	19.643	0.119	170.942	1.679	7.1185	0.3289	0.5014	0.0140	1.965E-16	13.64	5.81	20.59	7.24	0.060
1400	73.849	6.755	18.376	0.055	170.537	1.186	6.1620	0.4960	0.3298	0.0227	1.838E-16	43.12	5.45	64.13	11.53	0.069
EC-8[c]																
450	0.752	0.061	0.470	0.003	1.967	0.020	0.2997	0.0089	0.0041	0.0002	4.760E-18	38.22	3.19	0.73	0.12	1.434
500	3.402	0.846	8.896	0.062	8.284	0.092	0.2537	0.0224	0.0165	0.0029	8.924E-17	41.06	14.22	3.28	1.63	1.694
550	0.509	0.096	4.053	0.028	2.207	0.029	0.2458	0.0157	0.0057	0.0003	4.103E-17	23.05	24.53	0.49	0.18	1.749
600	0.374	0.032	1.755	0.017	1.028	0.015	0.3009	0.0128	0.0022	0.0001	1.801E-17	36.40	23.14	0.36	0.06	1.429
650	0.459	0.028	1.601	0.015	1.334	0.018	0.3808	0.0220	0.0030	0.0001	1.633E-17	34.39	16.16	0.44	0.05	1.129
700	0.586	0.049	1.009	0.008	1.826	0.021	0.6190	0.0214	0.0043	0.0002	1.024E-17	32.10	7.40	0.57	0.09	0.694
750	1.216	0.141	0.726	0.006	2.709	0.030	0.7814	0.0625	0.0052	0.0005	7.328E-18	44.86	3.57	1.17	0.27	0.550
850	1.219	0.159	0.881	0.008	4.384	0.051	0.9184	0.0489	0.0109	0.0005	8.862E-18	27.79	2.67	1.18	0.31	0.468
1000	2.403	0.618	2.060	0.037	7.946	0.174	1.5295	0.1193	0.0191	0.0020	2.067E-17	30.22	3.43	2.32	1.19	0.281
1150	2.489	0.732	1.133	0.009	11.124	0.122	4.8063	0.1862	0.0305	0.0025	1.136E-17	22.30	1.34	2.40	1.41	0.089
1350	11.215	1.162	0.920	0.006	34.609	0.409	7.2172	0.3139	0.0812	0.0039	9.212E-18	32.25	0.35	10.79	2.23	0.059

[a] Ages were calculated using GA1550 biotite with an age of 98.79 ± 0.96 Ma and 40K decay constant of 5.543×10^{-10} yr^{-1}. Correction factors for interfering neutron reactions were determined by analyzing K-glass and optical grade CaF_2: $(^{40}Ar/^{39}Ar)_K = 2.66 \times 10^{-2}$, $(^{36}Ar/^{37}Ar)_{Ca} = 2.70 \times 10^{-4}$, and $(^{39}Ar/^{37}Ar)_{Ca} = 6.76 \times 10^{-4}$. Data were corrected for line blanks determined by measurements at each temperature, discrimination was calculated by measurements of atmospheric argon (1.0135 ± 0.5%), and ^{37}Ar and ^{39}Ar decay after irradiation. Errors in summary age = 2σ and include error in J.

[b] J = 0.00049 ± 0.5%. Plateau age = 1.89 ± 0.10 Ma, MSWD = 1.76, steps 4–6.
[c] J = 0.000535 ± 0.5%. Plateau age = 0.45 ± 0.08 Ma, MSWD = 4.5, steps 3–6.
[d] J = 0.000490 ± 0.5%. Plateau age = 1.77 ± 0.24 Ma, MSWD = 0.73, steps 6–10.
[e] J = 0.000490 ± 0.5%. Plateau age = 1.51 ± 0.26 Ma, MSWD = 2.59, steps 4–7.
[f] J = 0.000487 ± 0.5%. Plateau age = 1.46 ± 0.05 Ma, MSWD = 1.98, steps 5–10.
[g] J = 0.00049 ± 0.5%. Plateau age = 2.59 ± 0.47 Ma, steps 3–7.
[h] J = 0.00049 ± 0.5%. Plateau age = 2.15 ± 0.19, steps 4–9.
[i] J = 0.000435 ± 0.5%. Error plateau age = 0.44 ± 0.15 Ma, MSWD = 9.37, steps 2–11.
[j] J = 0.000435 ± 0.5%. Plateau age = 2.71 ± 0.19 Ma, MSWD = 2.63.
[k] J = 0.00049 ± 0.5%.Plateau age = 1.33 ± 0.30 Ma, MSWD = 0.10, steps 1–3.
[l] J = 0.00049 ± 0.5%. Plateau age = 1.67 ± 0.19 Ma, MSWD = 4.71, steps 1–6.
[m] J = 0.00049 ± 0.5%. Plateau age = 2.44 ± 0.20 Ma, MSWD = 3.20, steps 1–9.
[n] J = 0.00049 ± 0.5%. Plateau age = 1.78 ± 0.09 Ma, MSWD = 4.10, steps 3–5.
[o] J = 0.00049 ± 0.5%. Plateau age = 0.21 ± 0.03 Ma, MSWD = 0.33, steps 5–7.

Table 4. Summary Age Information for Lavas[a]

Sample	Plateau Age ± 2σ, Ma	Steps on Plateau	MSWD Plateau	Isochron Age ± 2σ, MA	MSWD Isochron	$^{40}Ar/^{36}Ar$ Initial Ratio of Isochron	Comments or Preferred Age
EC-6	1.89 ± 0.10	4 to 6	1.76	1.92 ± 0.31	3.45	288 ± 111	plateau age
EC-8	0.45 ± 0.08	3 to 6	4.5	0.22 ± 15	1.21	372 ± 61	error plateau = maximum age
EC-9	1.77 ± 0.24	6 to 10	0.73	0.04 ± 0.08	0.34	365 ± 124	plateau age
EC-13	1.51 ± 0.26	4 to 7	2.59	0.65 ± 0.63	1.19	371 ± 88	plateau age
EC-14	1.46 ± 0.05	5 to 10	1.98	1.47 ± 0.09	2.44	296 ± 24	plateau age
EC-16	2.59 ± 0.47	3 to 7	1.7	2.19 ± 1.66	2.22	313 ± 120	plateau age
EC-17	2.15 ± 0.19	4 to 9	1.31	1.81 ± 0.37	0.4	324 ± 28	plateau age
EC-27	0.44 ± 0.15	2 to 11	9.37	ND	5.6	437 ± 145	plateau age
EC-29	2.71 ± 0.19	3 to 8	2.63	2.57 ± 0.86	3.22	317 ± 175	error plateau
EC-47	1.33 ± 0.30	1 to 3	0.1	1.93 ± 2.88	0.05	271 ± 116	plateau age
EC-51	1.67 ± 0.19	1 to 6	4.71	1.85 ± 0.24	3.23	275 ± 21	plateau age
EC-62	2.44 ± 0.20	1 to 9	3.2	2.24 ± 0.29	2.47	313 ± 20	plateau age
EC-69	1.78 ± 0.09	3 to 5	4.1	1.52 ± 0.43	0.03	898 ± 1078	plateau age
EC-70	0.21 ± 0.03	5 to 7	0.33	0.25 ± 0.12	0.29	248 ± 136	plateau age

[a] Location data are given in Table 1. ND indicates no data.

transitional. The VADM values range from 1.39 × 10^{22} Am^2 to 22.6 × 10^{22} Am^2. The average value for the normal sites is 10.2 10^{22} Am^2 (n = 5) and 3.2 10^{22} Am^2 (n = 8) for the reverse sites. The overall value for the combined mean is 6.0 × 10^{22} Am^2.

6. The $^{40}Ar/^{39}Ar$ Data

[18] Samples were analyzed in the $^{40}Ar/^{39}Ar$ laboratory at the Department of Geological Sciences at the University of Florida. Fresh groundmass concentrates weighing 50–100 mg were wrapped in Al foil and loaded into a quartz glass tube along with 1 mg packages of the flux monitor GA1550 biotite. Flux monitor packages were placed between every two basalt samples in the quartz tube. The samples and flux monitors were irradiated at the Oregon State reactor facility. Samples were degassed using a double vacuum resistance furnace attached to a stainless steel extraction and clean-up line. Reactive gasses were removed with SAES getters prior to expansion to the mass spectrometer. Argon

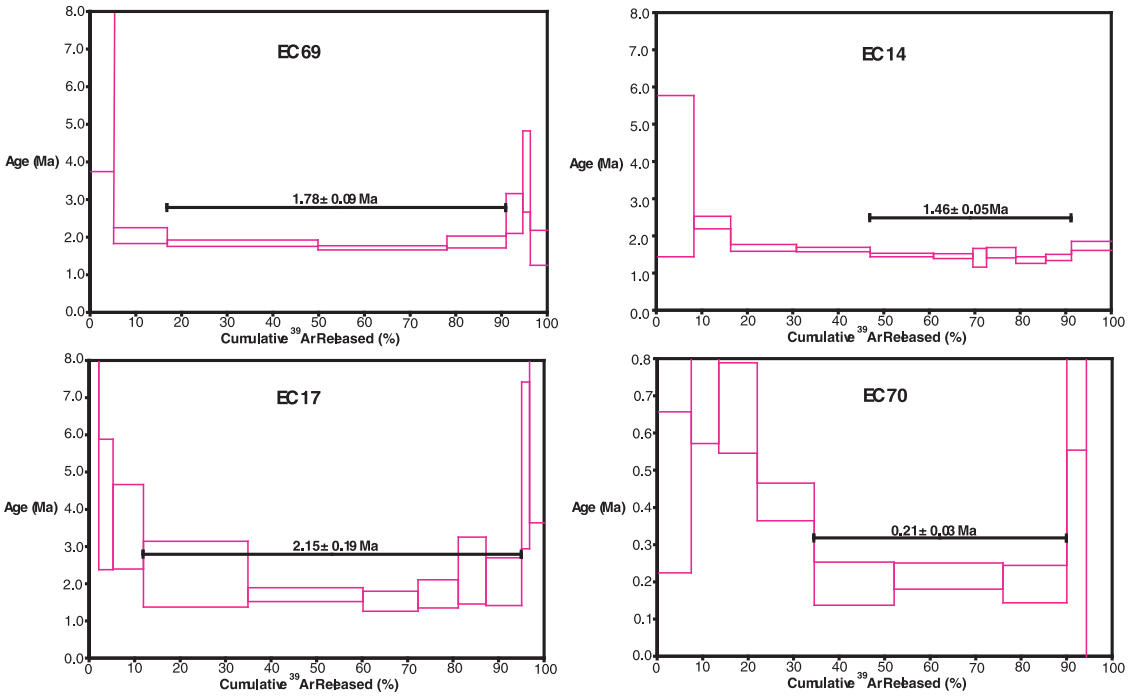


Figure 6. Examples of age spectra diagrams.

isotopes were analyzed using a MAP215-50 mass spectrometer with a Balzers electron multiplier. The data were reduced using ArArCALC software [*Koppers*, 2002], and apparent ages were calculated assuming an age of 98.79 ± 0.96 Ma for GA1550 biotite [*Renne et al.*, 1998].

[19] The analytical data for all of the samples is given in Table 3, and summary age information is given in Table 4. Figure 6 gives examples of age spectra diagrams for selected samples. The samples gave $^{40}Ar/^{39}Ar$ ages ranging from 2.59 ± 0.47 to 0.21 ± 0.03 Ma (±2 sigma errors). Most samples yielded plateau ages for significant fractions of the gas released (Table 4 and Figure 6). Inverse isochrons of the steps used for the age calculations show $^{40}Ar/^{36}Ar$ intercepts within error of atmosphere for all but one sample (EC-8). Some samples gave saddle-shaped age spectra suggesting the presence of excess argon in low- and high-temperature steps. Excess argon does not, however, appear to affect the steps used for age calculations except for possibly sample EC-8 (Table 4).

7. Magnetic Polarity Stratigraphy

[20] The road from Pifo to Papallacta bisects the Chacana Volcanic Complex which is capped by normal magnetized flows, the most recent being a flow that erupted in 1760 and is site 20 on Figure 1 and Table 1. The sites in order of age are shown in Figure 7 based on isotopic dating and superposition. The change from normal to reverse polarity occurs between sites 6 and 8 and represents the change from Brunhes to Matuyama age flows. The normal sites are Brunhes in age. A series of sites were studied along the south wall of the canyon that cuts into the west flank of the Chacana Volcanic Complex. All of the flows in this stratigraphic sequence have reverse magnetization, the upper flow of which has a K/Ar date of about 0.98 Ma [*Barberi et al.*, 1988]. The basal flow of the sequence (site 29) yields a date of 2.71 ± 0.19 Ma, the oldest rocks of the complex. Samples were taken on the east flank of the complex where the volcanic sequence is in contact with the underlying metamorphic rocks and these sites have reverse magnetization. The Chacana Volcanic Complex is therefore of Matuyama/Brunhes age.

8. Time-Averaged Field

[21] The present-day magnetic field in Ecuador is directed almost due north but with an inclination on the equator of about 25°. This is a large

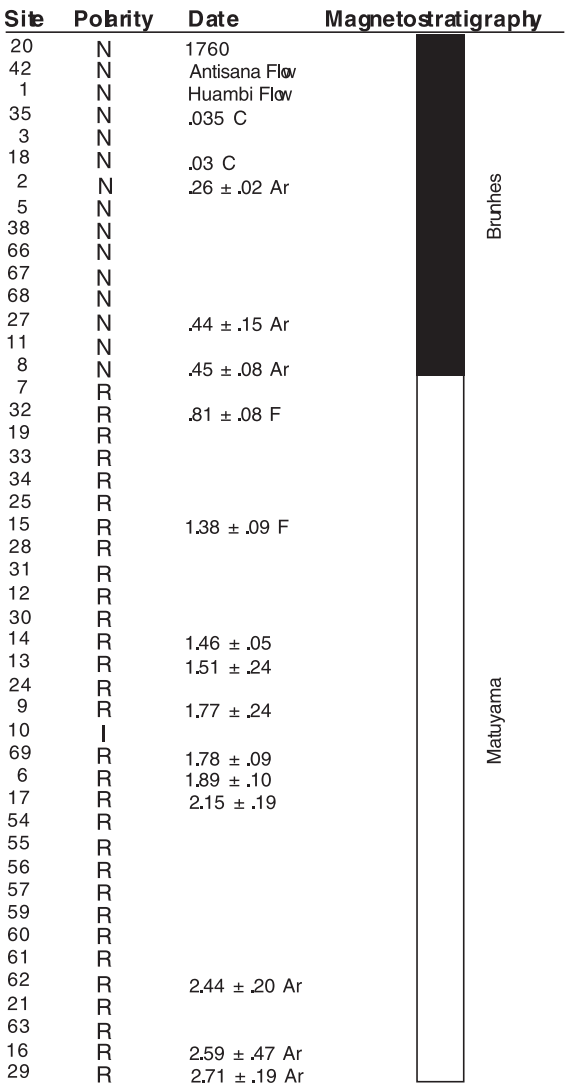


Figure 7. Magnetic stratigraphy of the Chacana Structure, Ecuador. All dates are in Ma except for the year 1760.

departure from the expected axial dipole inclination for this latitude and is the steepest observed inclination anywhere in the world at the equator. The magnetic field is also changing rapidly over the southern part of the continent because of the growth of negative flux patches at the surface of the core [*Johnson and Constable*, 1995; *Hulot et al.*, 2002]. However to the west of Ecuador secular variation is low [*Gubbins and Gibbons*, 2004]. The data presented in Table 1 rarely reach inclination values of 25°. The dispersion of the VGPs calculated from the site mean directions around the present axis of rotation for the selected data yields an ASD of 13.3°, which is not significantly different from model "G" of *McFadden et al.* [1991].

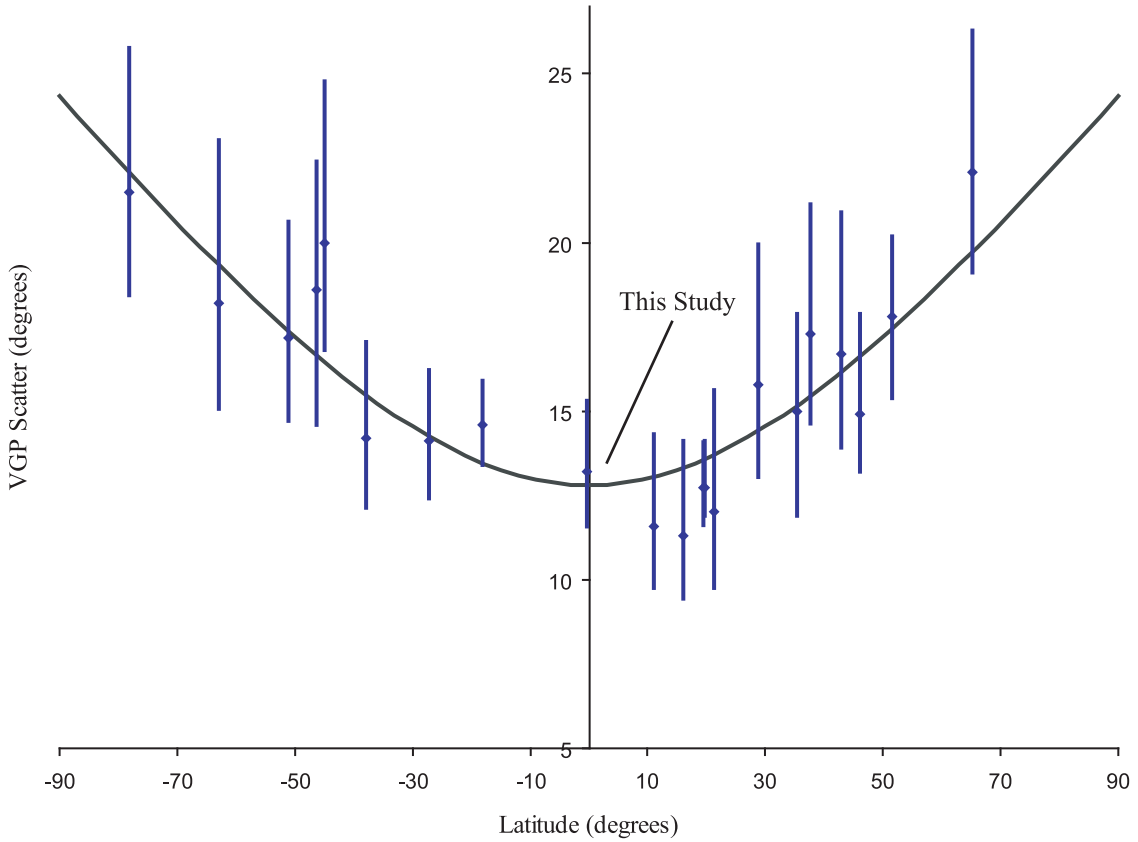


Figure 8. ASD of VGPs for recent studies from *Opdyke and Mejia* [2004] with addition of data obtained in this study (latitude 0.37°S) and by *Brown et al.* [2004] in Patagonia (latitude 45° S), plotted against model G of *McFadden et al.* [1988].

Rochette et al. [1997] in a study of 79 lava flows from the Galapagos Islands report an ASD of 11.2° while *Cox* [1971] reports a value of 10.9° which is not significantly different from the value presented here. It can be concluded that Earth's field is not particularly noisy at the Equator when examined over the last several million years. Figure 8 presents the dispersion of the time averaged magnetic field with latitude from high-quality studies of VGP dispersion with the results of this study added [*Opdyke and Mejia*, 2004], the data fits very well model "G" of *McFadden et al.* [1988]. The data presented in Figure 8 is taken from studies meeting modern standards and we conclude that this model accurately describes the dispersion expected by Earth's magnetic field when averaged over times on the order of one million years.

[22] One recent study gave results that are significantly different from model G (Figure 8). This is a study by *Brown et al.* [2004] on lavas from northern Patagonia. The study is interesting since almost 20% of the data set gives intermediate or transitional directions. This percentage is higher than any recent study of the TAF. For instance, only 100 km to the south of the *Brown et al.* [2004] study, *Mejia et al.* [2004], in a larger study, only observed 1 intermediate direction. *Opdyke and Musgrave* [2004] in a study of comparable size to the *Brown et al.* [2004] in Australia, found only two intermediate sites out of 42. The *Brown et al.* [2004] study is therefore considered to be anomalous, and the reason for the large dispersion of the directions even after the transitional directions are eliminated is unclear.

9. Conclusions

[23] One of the reasons this study was undertaken was to determine if the quadrupole field which should be at a maximum at the equator, can be detected. The difference between inclination and the expected dipole field inclination (ΔI) for the normal directions is $-4.8°$, $\alpha_{95} = 7.5°$ and for the

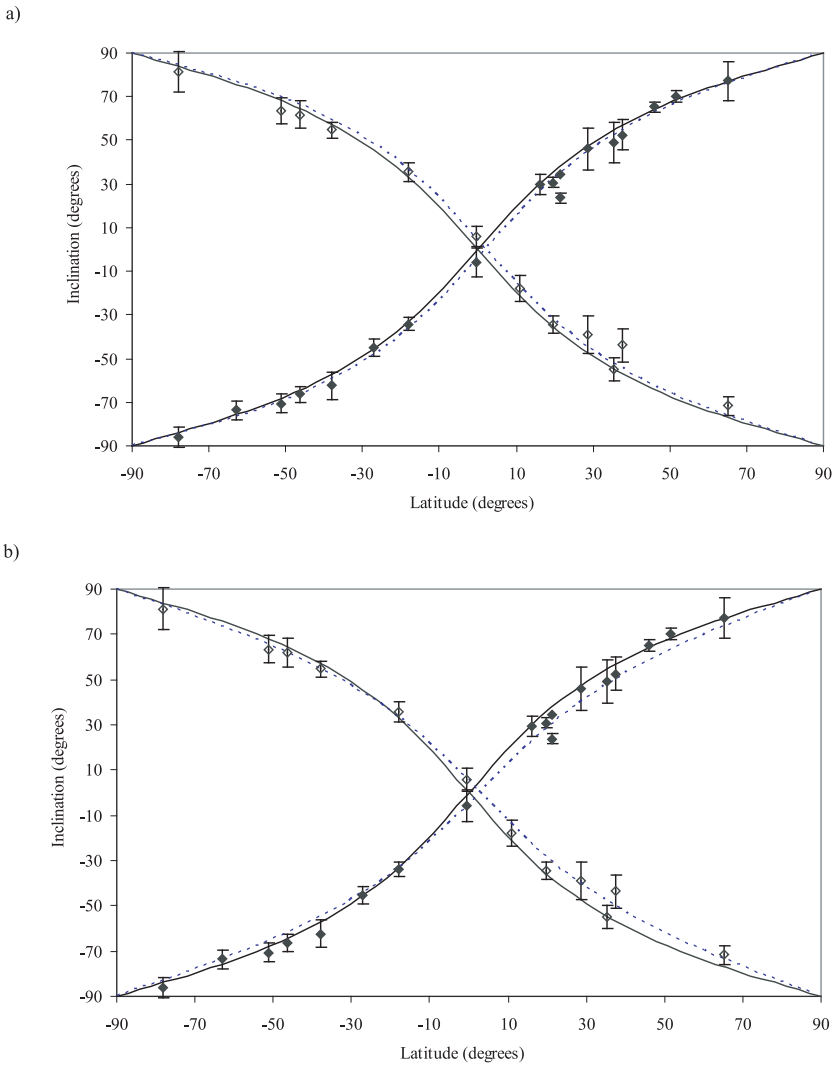


Figure 9. Latitude versus inclination for this study and recent studies of the time averaged field compiled by *Opdyke and Mejia* [2004]. Solid line represents the change with latitude of the geocentric axial dipole field, GAD. The dashed line represents the change with latitude of (a) the GAD + 5% quadrupole field and (b) the GAD + 5% quadrupole field + 7% octupole field.

reverse direction +4.9°, α_{95} = 4.9°. These values are essentially the same. The normal direction is not significantly different from the dipole field, however the reverse mean direction is significantly different, however, there are more reverse sites (30) than normal sites (21) which leads to a larger α_{95} for the normal sites. When the directions are combined, ΔI remains the same and the α_{95} is reduced to 4.1°, making the mean direction just significantly different from the geocentric axial dipole field. This amount of deviation from the geocentric axial dipole is best fit to a magnetic field dominantly dipolar with the addition of a 5% quadrupole field (Figure 9). This result seems to indicate that the Time Averaged Paleomagnetic Equator is displaced by several degrees into the Northern Hemisphere as originally predicted by *Wilson* [1970]. Although this effect is small it is significant and may only be possible to detect near the geographic equator where contributions from the quadrupole field will be most apparent. The value is consistent with other workers whose conclusions were based on the analysis of large data sets [*Merrill et al.*, 1996]. This observation must be confirmed and if corroborated implies a basic asymmetry in Earth's magnetic field. The declina-

tion of the combined data set is 359.9° giving no declination anomaly (ΔD) at all and provides a strong argument against tectonic rotations in the region of study over the last 2.6 Ma.

[24] There is no apparent difference between the normal and reverse field in this study with the directional values being the same for each data set. Unlike many studies of the time averaged field this study has more reverse flows than normal flows. Looking at the data set that is presented in Figure 9 it is apparent that the reversed localities are systematically too shallow relative to GAD. However, in most cases the means are not significantly different from the GAD field. There are no systematic departures from GAD in the site means of the normal sites. The reverse data set is best modeled by a field that is composed of the dominant GAD with 5% quadrupole field and in addition 7% octupole field (Figure 9b). *Johnson et al.* [2005] came to a similar conclusion when examining the TAFI data set.

[25] The data obtained in this study on the dispersion of the time averaged field is in agreement with model "G" of *McFadden et al.* [1988]. The present magnetic field is very divergent from a dipole field in Ecuador and in the Southern Hemisphere in general and noisier than expected when compared to the averaged paleomagnetic field. This has led *Opdyke and Mejia* [2004] to propose that Earth's magnetic field is becoming transitional and may be a precursor for a coming excursion or reversal.

[26] This study is the first on equatorial lavas to yield a significant number of successful paleointensity results. There has been considerable discussion about the average intensity of Earth's magnetic field and some believe that the present magnetic field is anomalously high [*Juarez and Tauxe*, 2000]. The paleointensity studies indicate that the average value of the VADM in Ecuador is 6.00×10^{22} Am^2, similar to the one obtained by *Juarez and Tauxe* [2000] of 5.5×10^{22} Am^2.

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